

Nathan Zhao

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Education

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Electrical Engineering (Circuit Technology and Robotics)

Expected May 2029

The Pembroke Hill School

Kansas City, Missouri

High School Diploma, Cum Laude

- **Honors & Coursework:** ACT 36 · GPA 97.7 · Multivariable Calculus, Differential Equations

Work Experience

Machine Learning Engineer · Effluent

2026 – Present

- Designed ML architecture fusing sparse attention with dilated inception graph convolutional networks for time-series forecasting of wastewater plant dynamics, optimized for per-facility transfer learning.
- Built simulation tools, Python data pipelines, and a cross-platform C#/.NET GUI delivering actionable insights interpreted from XGBoost/Prophet forecasting directly to plant operators.

Research & Development Intern · MRIGlobal

2025 – 2026

- Led internal R&D project on developing custom chemically-coated gas sensor arrays for real-time detection and classification of chemical warfare agents. Submitted to DoD DTRA CBOA.
- Designed and fabricated a custom STM32F411CEU6 PCB (Altium) and deployed a CNN-BiLSTM model on-device (TensorFlow Lite), reaching >96% classification accuracy on edge inference.

Research Assistant · UMKC Multiplatform Interactive Robotics Lab (mirolab)

2025 – 2026

- Led development of vision-based hand-joint goniometry system fusing multi-view MediaPipe hand pose with FDM 3D-printed ArUco marker glove via nonlinear Kalman filtering, reaching flexion-angle accuracy within 0.8°.
- Co-authored paper, to submit to IEEE RA-L (PI: Hee Sup Shin).

Research Intern · University of Pittsburgh, Accessible SmartTech Lab

2024

- Programmed autonomous object-handoff behaviors on a Hello Robot Stretch RE1 (ROS Noetic) using ArUco markres, YOLO object detection, and 2D LiDAR SLAM (PI: Dan Ding)

Selected Projects

Bimanual Mobile Manipulator for Assisted Living

2024 – Present

- Dual telescopic-arm mobile robot for in-home elderly-care tasks. *Onshape, C++, C#, PlatformIO.*

mmHome: Multimodal IoT Contactless Vital-Sign Monitoring & Fall Detection

2025 – 2026

- Wall-mounted mmWave + vibration sensing for respiration and fall detection. *Python, scikit-learn, C++.*

Multi-Agent Boids Swarm Control

2026 – Present

- Real-time MPC of Reynolds flocking parameters for coordinated robot navigation. *PyBullet, PyTorch.*

LeSCARA: Open Source SCARA Platform for Embodied AI Research

2026 – Present

- Low-cost 3D-printed SCARA arm for reproducible learning-based manipulation. *Onshape, C++, LeRobot.*

Relevant Skills

Computer Aided Design: Onshape, KiCAD, Altium Designer, SolidWorks

Programming: C++, Python, C#/.NET, MATLAB, SystemVerilog, $\LaTeX$, Linux, Docker, Git

ML & Vision: PyTorch, scikit-learn, OpenCV, TensorFlow Lite, XGBoost, pandas

Robotics & Embedded: ROS/ROS2, MuJoCo, PyBullet, PlatformIO, PicoSDK, Arduino

Languages: English, Chinese (Mandarin), Spanish